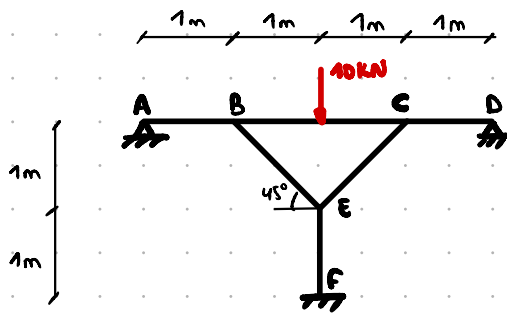


P.3.3.

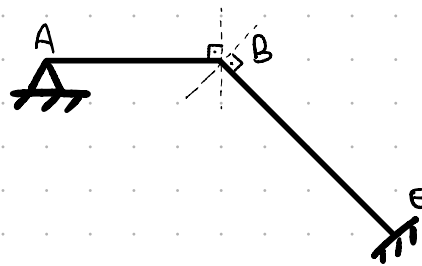
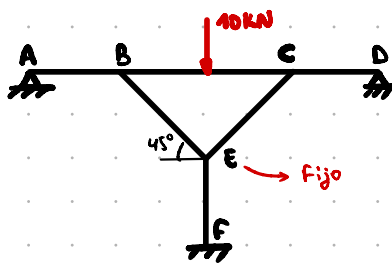


$$EI = 1 \cdot 10^3 \text{ KN}\cdot\text{m}^2$$

- GHE, TRAS/ INTRAS
- Flectores en CD y BC
- Axiles EF
- Movimientos en A y C

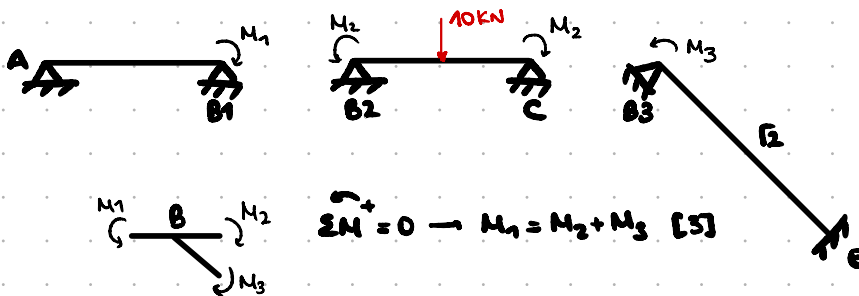
$$a) GHE = LE - GLE = 7 - 3 = 4$$

Todo lo que ocurre en el eje de simetría tiene que guardar esta simetría, por eso la reacción horizontal y el momento en el punto F son cero, de la misma forma todos los puntos en el eje no pueden desplazarse horizontalmente ni tener rotación.



Los nudos del punto B solo pueden moverse en sus perpendiculares, el único punto en el que ambos coinciden es en el cero, entonces no se desplazan **Ojo!! Sí que pueden rotar.**

Estructura intraslacional, separamos barras e igualamos giros:



$$\vec{\theta}_B = \vec{\theta}_B \quad [1]$$

$$\vec{\theta}_B = \vec{\theta}_B \quad [2]$$

$$\sum M = 0 \rightarrow M_1 = M_2 + M_3 \quad [3]$$

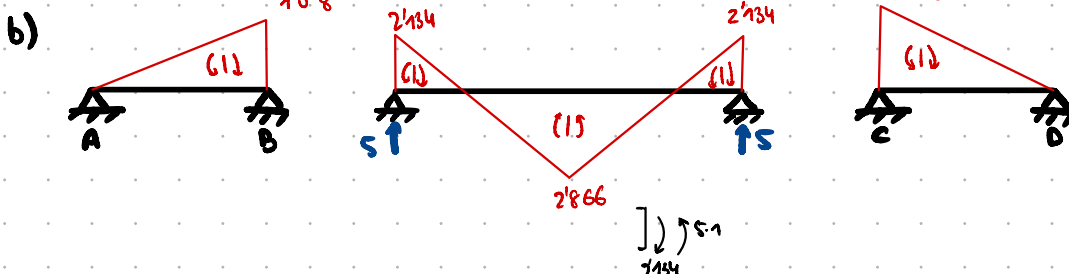
Las ecuaciones de los giros las saco del formulario.

$$[1] \rightarrow \frac{M_1 \cdot 1}{3EI} = - \frac{M_2 \cdot 2}{3EI} - \frac{M_2 \cdot 2}{6EI} + \frac{10 \cdot 1 \cdot 1}{6EI \cdot 2} \cdot (2+1)$$

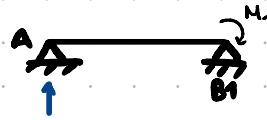
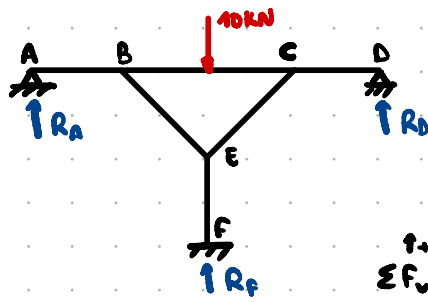
$$[2] \rightarrow \frac{M_1 \cdot 1}{3EI} = - \frac{M_3 \cdot \sqrt{2}}{4EI}$$

Resuelvo el sistema de 3 ecuaciones y 3 incógnitas con la calculadora

$$M_3 = -1'036 \text{ KN}\cdot\text{m} ; M_2 = 2'134 \text{ KN}\cdot\text{m} ; M_1 = 1'098 \text{ KN}\cdot\text{m}$$



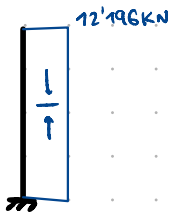
c) Hago equilibrios de verticales



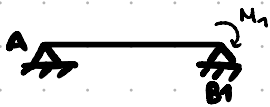
$$\sum \vec{M}_{B1} = 0 \rightarrow R_A \cdot 1 + 1098 = 0$$

$$R_A = -1098 \text{ KN} = R_D$$

$$\sum F_v = 0 \rightarrow R_F = 10 + 1098 \cdot 2 \rightarrow R_F = 12196 \text{ KN}$$



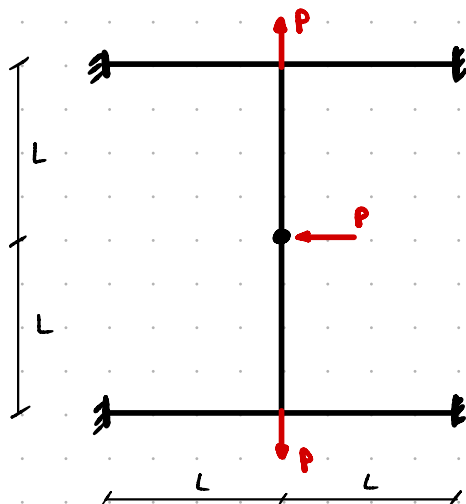
d) $\vec{\theta}_D$ y $\vec{\theta}_C$



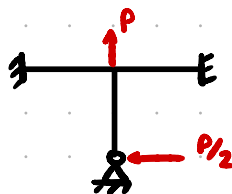
$$\vec{\theta}_A = -\frac{1098 \cdot 1}{6EI} = -1'83 \cdot 10^{-4} \text{ rad}$$

$$\vec{\theta}_C = \vec{\theta}_B = \vec{\theta}_{B1} = \frac{1098 \cdot 1}{3EI} = 3'66 \cdot 10^{-4} \text{ rad}$$

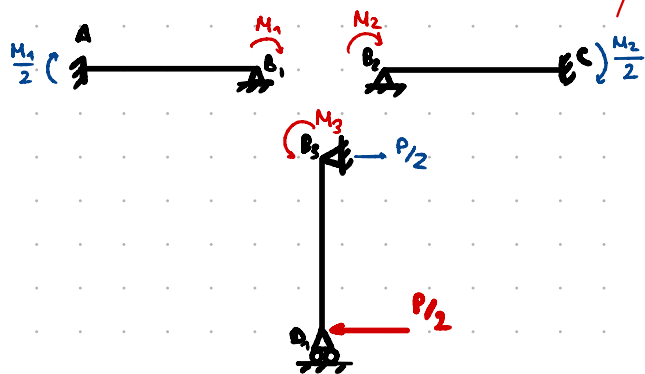
A) transnacional/intraslacional y simétrica/ antitética



Intraslacional
Simétrica



Separamos por tramos:



No aparece en el formulario,
hay que aprenderse!!

$$\sum \vec{M}_B = 0 \rightarrow M_3 - \frac{PL}{2} = 0 \rightarrow M_3 = \frac{PL}{2}$$

$$\begin{aligned} \text{[1]} \quad \vec{\theta}_B = \vec{\theta}_C = \vec{\theta}_D \\ \text{[2]} \quad M_1 + M_2 - M_3 = 0 \end{aligned}$$

Ecuación [1]

$$\frac{M_1 L}{4EI} = \frac{M_2 L}{4EI} \rightarrow M_1 = M_2$$

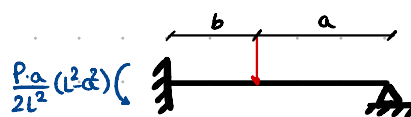
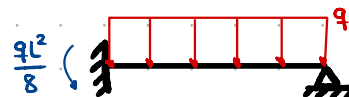
Ecuación [2]

$$M_1 + M_2 = \frac{PL}{2} \rightarrow \boxed{M_1 = M_2 = \frac{PL}{4}}$$

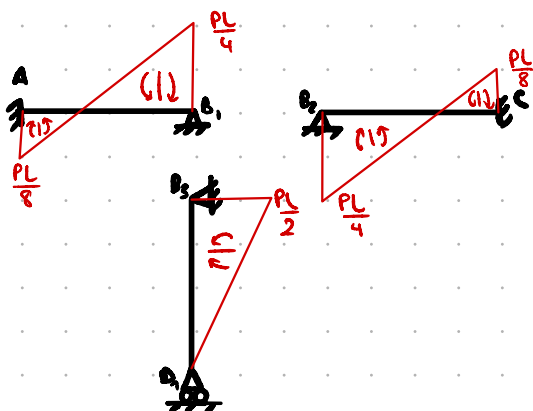
REACCIONES

$$\sum \vec{M}_A = 0 \rightarrow R_{B1} \cdot L = \frac{3M_1}{2} \rightarrow R_{B1} = \frac{3P}{8}$$

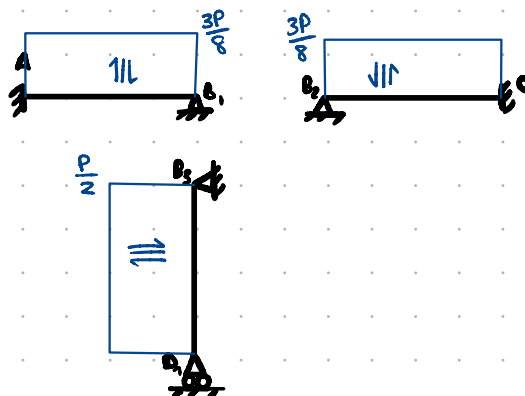
No formulario!! APRENDER



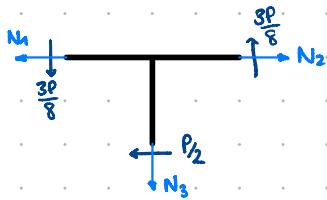
b) LEY FLECTORES



c) LEY CORTANTES



EQUILIBRIO NUDO B



$$\sum F_y = 0 \rightarrow -P - N_3 = 0 \rightarrow \boxed{N_3 = -P}$$

d) Movimientos en D: $\bar{u}_0$; $\bar{v}_0$; $\bar{\theta}_0$

$$\boxed{\bar{v}_0 = 0}$$

$$\boxed{\bar{u}_0 = \bar{\theta}_{B3} \cdot L + \frac{1}{EI} \cdot \left[\frac{1}{2} \cdot \frac{P}{2} \cdot \frac{2L}{3} \right] = \frac{11}{48} \cdot \frac{P L^3}{EI}}$$

$$\boxed{\bar{\theta}_0 = \bar{\theta}_{B3} + \frac{1}{EI} \cdot \left[\frac{1}{2} \cdot \frac{P L}{2} \right] = \frac{5}{16} \cdot \frac{P L^2}{EI}}$$

GIRO RELATIVO

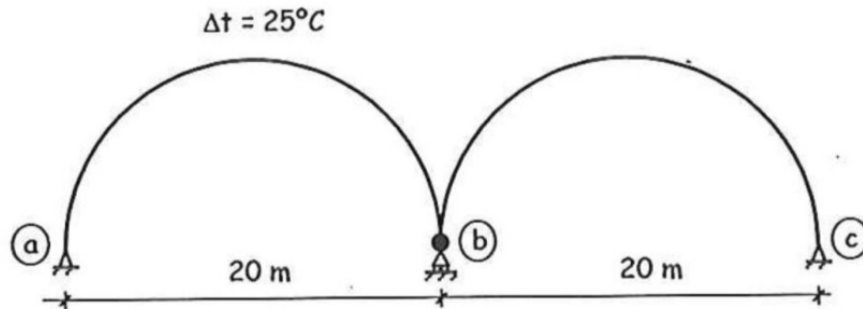


$$\boxed{\theta_{RELAT} = 2 \cdot \theta_{D1} = \frac{10}{16} \cdot \frac{P L^2}{EI}}$$

PROBLEMA 3.8

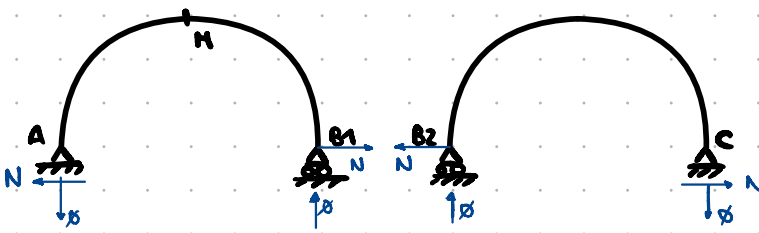
La estructura de la figura está formada por dos arcos semicircular iguales cuya rigidez es $EI=800\,000\text{ kNm}^2$. La unión de los arcos es una rótula que permite la existencia de giro relativo entre las piezas. La única carga actuante es un incremento de temperatura de 25°C en el arco izquierdo. El coeficiente de dilatación térmico de ambos es de $10^{-5}\text{ }^\circ\text{C}^{-1}$. Se pide:

- 1) Explicar razonadamente cual es el procedimiento a seguir para resolver el problema.
- 2) Reacciones
- 3) Ley de momentos flectores, axiles y cortantes en la estructura.
- 4) Movimientos horizontal y vertical de la clave del arco izquierdo.



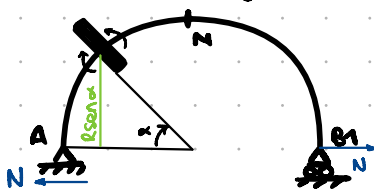
1) PROCEDIMIENTO RESOLUCIÓN

$GH = 5 - 3 - Rot = 1$ — Cortamos por la rótula e igualamos desplazamientos de B



CÁLCULO DE $\bar{u}_{B1}$ y $\bar{u}_{B2}$

$$\bar{u}_{B1} = \frac{1}{EI} \cdot \int_0^\pi \overbrace{NR \sin \alpha}^{M(x)} \cdot \underbrace{R \sin \alpha}_{\text{Distancia vertical a B1}} \cdot \underbrace{R d\alpha}_{ds} + \alpha \Delta T \cdot 2R = \frac{NR^3}{EI} \cdot \int_0^\pi \sin^2 \alpha d\alpha + \alpha \Delta T \cdot 2R = \frac{N \cdot 10^3}{8 \cdot 10^5} \cdot \frac{\pi}{2} + 10^{-5} \cdot 25 \cdot 20$$



$$M(x) = NR \sin \alpha$$

$$\bar{u}_{B2} = -\frac{1}{EI} \cdot \int_0^\pi NR \sin \alpha d\alpha \cdot R \sin \alpha \cdot R d\alpha = -\frac{NR^3}{EI} \cdot \frac{\pi}{2} = -\frac{N \cdot 10^3 \pi}{8 \cdot 10^5 \cdot 2}$$

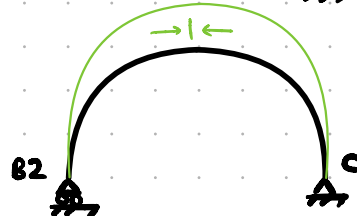
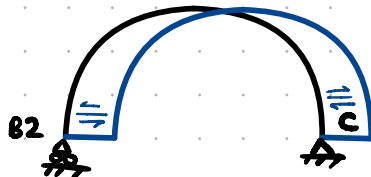
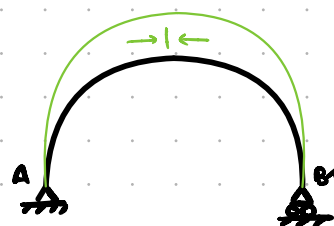
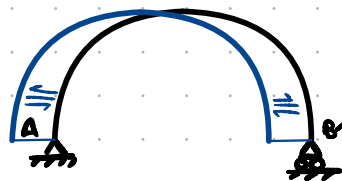
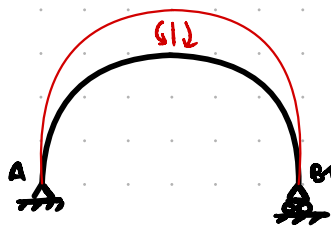
$$\bar{u}_{B1} = \bar{u}_{B2} \rightarrow \boxed{N = -1'2755 \text{ kN}}$$

3) LEYES M, N, Q

$$M(\alpha)_{A_{B1}} = -12'755 \sin \alpha$$

$$Q(\alpha) = N \cdot \cos \alpha$$

$$N(\alpha) = N \sin \alpha$$



$$\dot{V}_0 = 0 = \bar{\Theta}_A 2R + \frac{1}{EI} \left[- \int_0^{\pi/2} M(\alpha) R (\cos \alpha + 1) R d\alpha \right] + \text{La temperatura no aporta al movimiento vertical}$$

$$\rightarrow \boxed{\bar{\Theta}_A = 159'44 \cdot 10^{-6} \text{ rad}}$$

$$\bar{u}_M = \bar{\Theta}_A R + \frac{1}{EI} \left[\int_0^{\pi/2} M(\alpha) R (1 - \sin \alpha) R d\alpha \right] + \alpha \Delta T R \rightarrow \boxed{\bar{u}_M = 56'34 \cdot 10^{-5} \text{ m}}$$

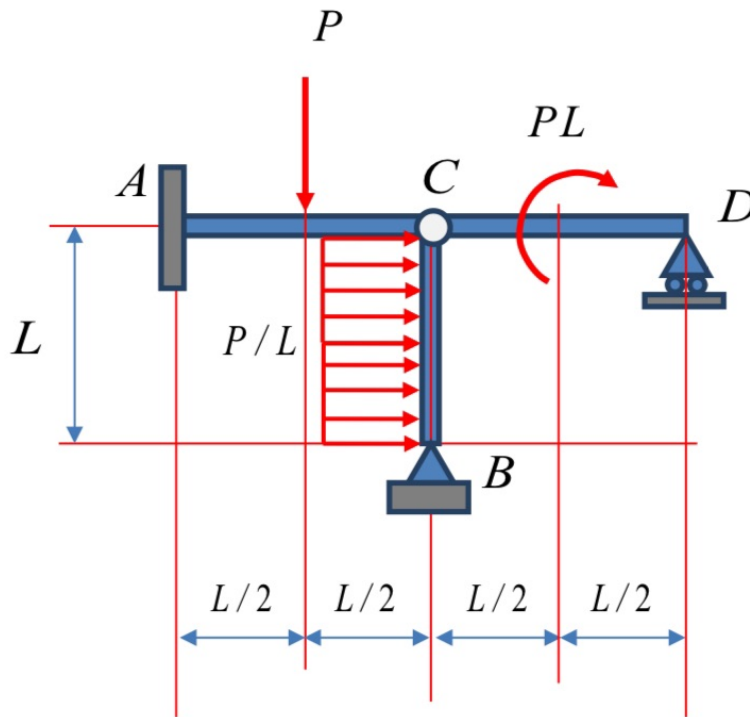
$$\dot{V}_M = \bar{\Theta}_A R + \frac{1}{EI} \int_0^{\pi/2} -M(\alpha) R \cos \alpha \cdot R d\alpha + \alpha \Delta T R \rightarrow \boxed{\dot{V}_M = 3'3 \cdot 10^{-3} \text{ m}}$$

$$\bar{\Theta}_M = \bar{\Theta}_A + \frac{1}{EI} \int_0^{\pi/2} M(\alpha) R d\alpha$$

PROBLEMA 3.10

Para la estructura de la figura, se pide calcular en función de la rigidez EI :

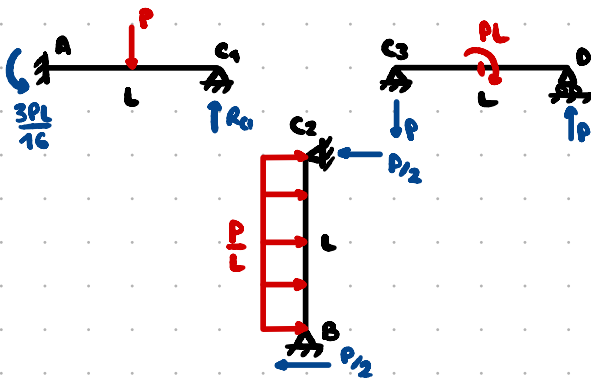
1. Indicar el grado de hiperestatismo de la estructura; si la estructura es traslacional o intraslacional y si se puede simplificar utilizando simetrías o antimetrías. Explicar brevemente los pasos a seguir para solucionarla.
2. Calcular las reacciones en A, B y D.
3. Calcular los diagramas de esfuerzos en la barra AC.
4. Calcular el giro relativo entre las barras AC y CB.
5. Calcular el movimiento del punto medio de la barra AC.



1) INTRASLACIONAL

$$GH = 6 - 3 - 2 = 1$$

Separamos por tramos



Primer tramo del formulario

$$M_A = \frac{P L/2}{2 L^2} \left(L^2 - \left(\frac{L}{2} \right)^2 \right) = \frac{3PL}{16}$$

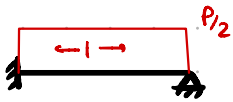
$$\sum \hat{M}_A = 0 \rightarrow M_A + R_C \cdot L - P \frac{L}{2} = 0 \rightarrow R_C = \frac{5P}{16}$$

Equilibrio en el nudo C

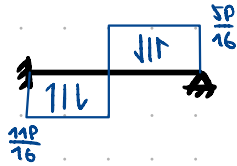
$$\begin{aligned} \frac{5P}{16} + N_2 &= P \rightarrow N_2 = \frac{11}{16} P \\ N_1 &= P/2 \end{aligned}$$

$$\sum F_v = 0 \rightarrow R_C + R_A - P = 0 \rightarrow R_A = \frac{11P}{16}$$

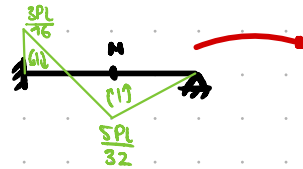
Diagramas en a-c



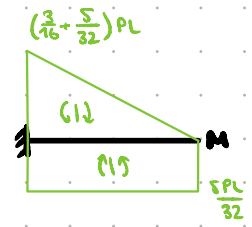
Axíl



Cortante



Momento flector



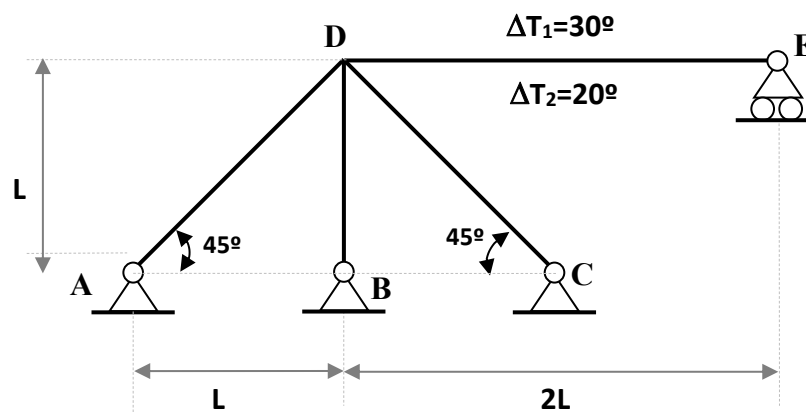
$$\delta_{VM} = \frac{1}{EI} \left[\frac{l}{2} \cdot \frac{1}{2} \left(\frac{3}{16} + \frac{5}{32} \right) PL \cdot \frac{2}{3} \cdot \frac{l}{2} - \frac{l}{2} \cdot \frac{5}{32} \cdot PL \cdot \frac{l}{4} \right]$$

PROBLEMA 3.16

La estructura, representada en la figura, está apoyada en sus extremos A, B y C, y en el extremo E tiene un apoyo deslizante horizontal. En la barra DE está actuando una sollicitación térmica $\Delta T_1=30^\circ\text{C}$ en la cara exterior y $\Delta T_2=20^\circ\text{C}$ en la cara interior. Todas las barras de la estructura tienen la misma rigidez a flexión EI , coeficiente de dilatación térmica α y sección rectangular de canto c .

Se pide, en función de L :

- Indicar el grado de hiperestatismo de la estructura y el carácter traslacional o intraslacional de la misma.
- Movimientos del nudo D
- Movimientos del nudo E
- Reacción en el apoyo E
- Giro del nudo A.



Datos:

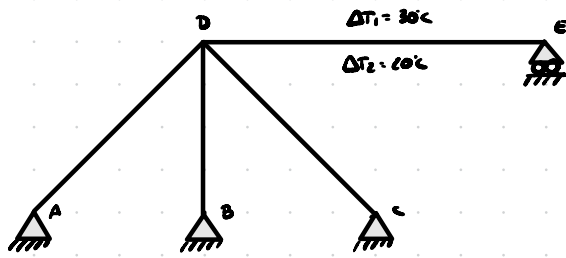
$$EI=5000L^2 \text{ kNm}^2$$

$$c=L/20$$

$$\alpha=10^{-5} \text{ } ^\circ\text{C}^{-1}$$

Nota:

Se despreciaran las deformaciones debidas a esfuerzos cortantes y axiales.



a) GM / Translational

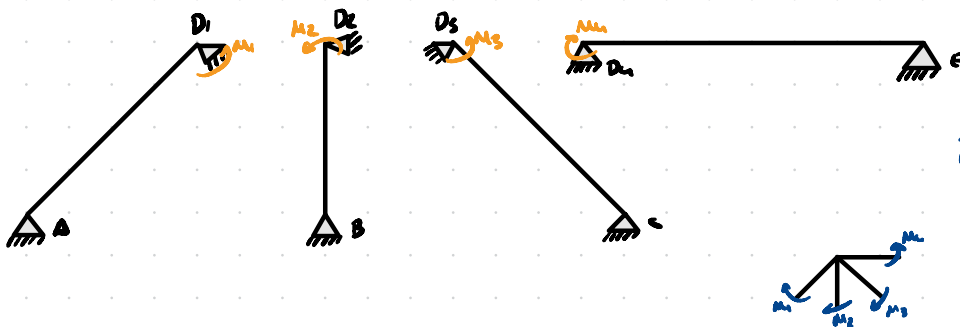
$$GM = 7 - 3 = 4$$

INTRANSLACIONAL

b)

$$\bar{u}_0 = \bar{v}_0 = 0$$

Separando barras



$$\bar{\theta}_{D1} = \bar{\theta}_{D2} = \bar{\theta}_{D3} = \bar{\theta}_{D4}$$

$$M_1 + M_2 + M_3 - M_4 = 0 \quad [1]$$

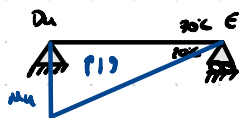
Cálculo de $\bar{\theta}_i$

$$\bar{\theta}_{D1} = \frac{M_1 \sqrt{2} L}{3EI}$$

$$\bar{\theta}_{D2} = \frac{M_2 L}{3EI}$$

$$\bar{\theta}_{D3} = \frac{M_3 \sqrt{2} L}{3EI}$$

Borra D4-E



$$\bar{v}_E = 0 = \bar{\theta}_{D4} \cdot 2L + \frac{1}{EI} \cdot \frac{1}{2} \cdot 2L \cdot M_4 \cdot \frac{2L}{3} - \alpha \frac{10}{1000} \cdot 2L \cdot L \rightarrow \bar{\theta}_{D4} = -\frac{M_4 \cdot 2L}{3EI} + 200 \cdot 10^{-5}$$

$$\bar{\theta}_{D1} = \bar{\theta}_{D2} \rightarrow M_1 \sqrt{2} - M_2 = 0 \quad [2]$$

$$\bar{\theta}_{D1} = \bar{\theta}_{D3} \rightarrow M_1 = M_3 \quad [3]$$

$$\bar{\theta}_{D1} = \bar{\theta}_{D4} \rightarrow M_1 \sqrt{2} + 2M_4 = 30L \quad [4]$$

$$\{ [1]; [2]; [3]; [4] \} \rightarrow \begin{cases} M_1 = 3,64L \\ M_2 = 5,15L \\ M_3 = 3,64L \\ M_4 = 12,43L \end{cases}$$

$$\bar{\theta}_0 = \bar{\theta}_{D1} = \frac{3,64 \sqrt{2} L}{3 \cdot 5 \cdot 10^3 L^2} = 3,43 \cdot 10^{-4} \text{ rad}$$

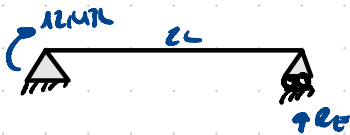
c) Momentos de E $1V_E = 0$

$$\bar{U}_E = \alpha \cdot \frac{30+20}{2} \cdot 2L = 50 \cdot 10^{-5} u$$

$$\bar{\theta}_E = \bar{\theta}_0 + \frac{1}{EI} \left[\frac{2L}{2} M_{u1} \right] - \alpha \cdot \frac{10}{L/10} \cdot 2L = 2,7 \cdot 10^{-5} \text{ rad}$$

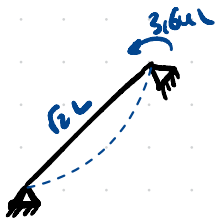
$\uparrow$ $\uparrow$ $\uparrow$
 $3,43 \cdot 10^3$ $5 \cdot 10^3 L^2$ $12,43L$

d) Reacciones en E



$$\sum M_{u1} = 0 \rightarrow R_F 2L - 12,43L = 0 \rightarrow R_F = 6,22 \text{ kN}$$

e) $\bar{\theta}_D$



$$\bar{\theta}_D = - \frac{3,6 \text{ kN} \cdot 2L}{6EI} = -1,72 \cdot 10^{-4} \text{ rad}$$

a) Transaccional o intraccional (1,5p)

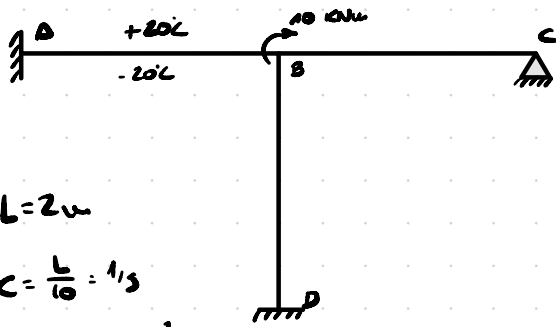
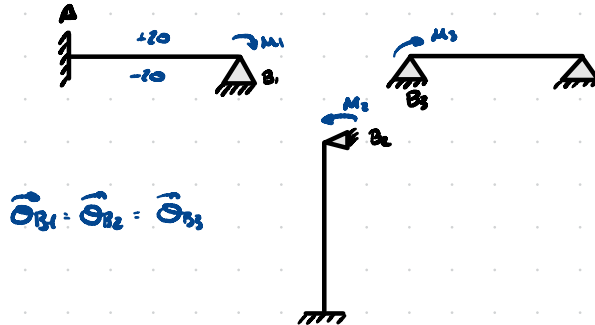
Intransaccional

$$L = 2u$$

$$C = \frac{L}{10} = 1/5$$

$$EI = 10000 \text{ kNm}^2$$

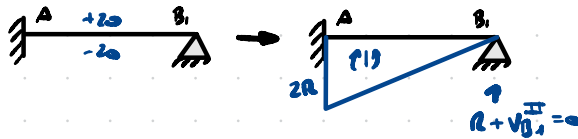
$$\alpha = 40^{-5} \cdot \text{rad}$$

b) $\bar{\theta}_B$ (4p)

$$M_1 - M_2 - M_3 = 0 \quad [1]$$

Cálculo de $\bar{\theta}_i$ $\bar{\theta}_{B1} \rightarrow$ separamos

$$\bar{\theta}_{B1}^I = \frac{M_1 \cdot L}{4EI}$$



$$1 V_B1^II = 0 = \frac{1}{EI} \cdot \left[\frac{2}{2} \cdot 2L \cdot \frac{2}{3} L \right] - \alpha \frac{40}{1/5} \cdot 2 \cdot 1 \rightarrow R = 4,5 \text{ kN}$$

$$\bar{\theta}_{B1}^{II} = - \frac{1}{EI} \cdot \left[\frac{2}{2} \cdot 2L \right] + \alpha \frac{40}{1/5} \cdot 2 = -0,001 \text{ rad}$$

$$\bar{\theta}_{B1} = \bar{\theta}_{B1}^I - \bar{\theta}_{B1}^{II} = \frac{M_1}{2EI} - 0,001$$

$$\bar{\theta}_{B2} = - \frac{M_2 \cdot L}{4EI}$$

$$\bar{\theta}_{B3} = - \frac{M_3 \cdot L}{3EI}$$

Ignorando:

$$\bar{\theta}_{B1} = \bar{\theta}_{B2} \rightarrow \frac{M_1}{2EI} - 0,001 = - \frac{M_2}{2EI} \rightarrow M_1 + M_2 = 2 \quad [2]$$

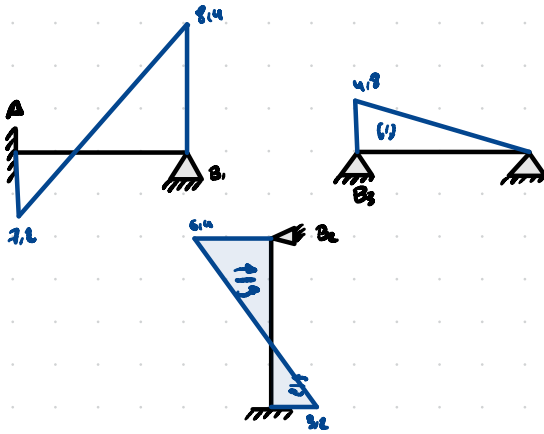
$$\bar{\theta}_{B2} = \bar{\theta}_{B3} \rightarrow - \frac{M_2}{2EI} = - \frac{M_3 \cdot L}{3EI} \rightarrow M_2 + \frac{4}{3} M_3 = 0 \quad [3]$$

Resolviendo [1][2], [3]

$$\begin{cases} M_1 = 8,4 \text{ kNm} \\ M_2 = -6,4 \text{ kNm} \\ M_3 = 4,8 \text{ kNm} \end{cases}$$

$$\bar{\theta}_B = - \frac{M_2}{2EI} = - \frac{6,4}{2000} = 0,0032 \text{ rad}$$

c)



d) ley de cortantes

A-B

$$R_{B1}^1 \cdot 2 = \frac{3,41}{2} \rightarrow R_{B1} = \frac{3,41}{4} = 0,85$$

$$R = 1,5$$

$$R_{B1}^{10} = 1,5 + 0,5 = 2,0$$

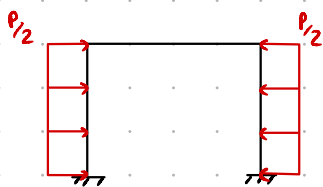
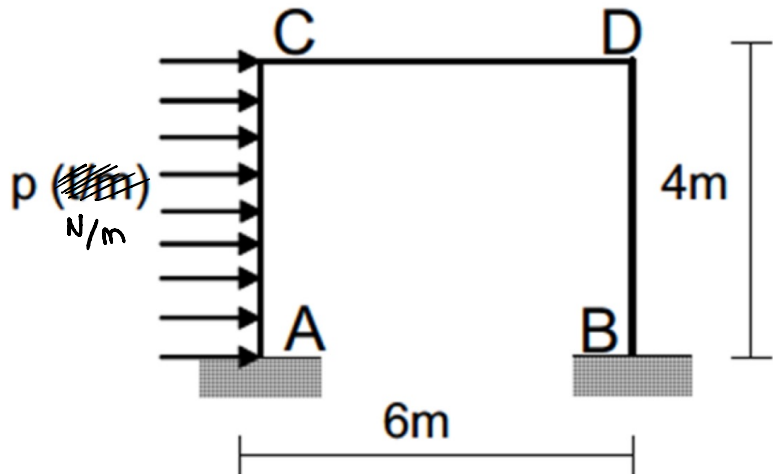
$$\underline{B-C} : M_3 = R_C \cdot 2 \rightarrow R_C = \frac{M_3}{2} = 7,4 \text{ kN}$$

$$\underline{B-D} : R_{D2} \cdot 2 = \frac{3M_2}{2} \rightarrow R_{D2} = \frac{3}{4}M_2 = 4,8 \text{ kN}$$

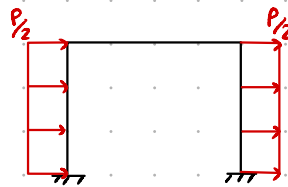
PROBLEMA 3.2

En la estructura de la figura y suponiendo las barras inextensibles con valor $EI=10800 \text{ N}\cdot\text{m}^2$ para la barra CD y $EI=3200 \text{ N}\cdot\text{m}^2$ para las barras AC y BD, utilizando la técnica de descomposición en una estructura simétrica y otra antisimétrica, calcule:

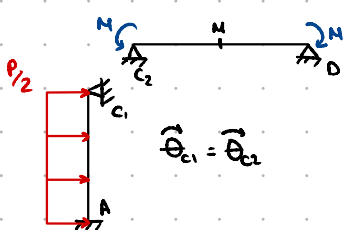
- El movimiento del punto medio del dintel
- Las reacciones en A.



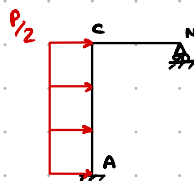
SIMÉTRICA
(INTRASLACIONAL)
cálculo: V_M



ANTISIMÉTRICA
(TRASLACIONAL)
cálculo: $\bar{\theta}_M$ y $\bar{u}_M$



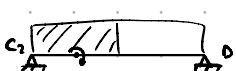
$$\bar{\theta}_{c1} = \bar{\theta}_{c2}$$



INTRASLACIONAL

$$\left. \begin{aligned} \bar{\theta}_{c1} &= \frac{M_c \cdot 4}{4 \cdot 32 \cdot 10^4} - \frac{P \cdot 4^3}{48 \cdot 32 \cdot 10^4} \\ \bar{\theta}_{c2} &= -\frac{M_c \cdot 6}{3 \cdot 108 \cdot 10^4} - \frac{M_c \cdot 6}{6 \cdot 108 \cdot 10^4} \end{aligned} \right\} \bar{\theta}_{c1} = \bar{\theta}_{c2} \rightarrow M_c = 0.553P$$

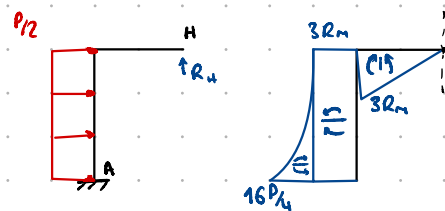
MOVIMIENTO $\bar{V}_M$



$$\bar{V}_M = \bar{\theta}_{c1} \cdot 3 + \frac{1}{108 \cdot 10^4} \left[-M_c \cdot 3 \cdot \frac{3}{2} \right] = \frac{P}{6800}$$

$$\bar{\theta}_{c2} = -\frac{P}{10700}$$

TRASLACIONAL



$$\bar{V}_M = \frac{1}{32 \cdot 10^4} \left[4 \cdot 3R_M \cdot 3 - \frac{1}{3} \cdot 4 \cdot 4 \cdot P \cdot 3 \right] + \frac{1}{108 \cdot 10^4} \left[\frac{1}{2} \cdot 3 \cdot 3R_M \cdot \frac{2}{3} \cdot 3 \right] = 0 \rightarrow R_M = \frac{12}{29}P$$

CÁLCULO DE $\bar{\theta}_M$ y $\bar{u}_M$

$$\bar{u}_M = \frac{1}{32 \cdot 10^4} \left[-4 \cdot 3R_M \cdot 2 + \frac{1}{3} \cdot 4 \cdot 4P \cdot \frac{3}{4} \cdot 4 \right] =$$

$$\bar{\theta}_M = \frac{1}{32 \cdot 10^4} \left[-4 \cdot 3R_M + \frac{1}{3} \cdot 4 \cdot 4P \right] + \frac{1}{108 \cdot 10^4} \left[-\frac{1}{2} \cdot 3 \cdot 3R_M \right] =$$

$$\rightarrow \bar{u}_M = \frac{11}{5800}P$$

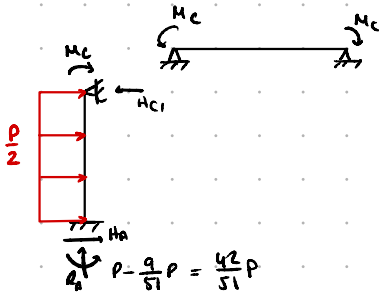
$$\bar{\theta}_M = -\frac{P}{17400}$$

REACCIONES

$$\sum F_H = 0 \rightarrow \frac{P}{2} \cdot 4 + H_A = 0 \rightarrow H_A = -2P$$

$$\sum F_V = 0 \rightarrow R_A + R_M = 0 \rightarrow R_A = -\frac{12}{29}P$$

$$\sum M_A = 0 \rightarrow M_A - 2P \cdot 2 + R_M \cdot 8 = 0 \rightarrow M_A = -\frac{80}{29}P$$



$$\sum M_A = 0 \rightarrow \frac{42}{51}P + H_{C1} \cdot 4 - M_C - 4P = 0 \rightarrow H_{C1} = \dots$$

$$\sum F_H = 0 \rightarrow H_A + \frac{P}{2} \cdot 4 - H_{C1} = 0 \rightarrow H_A = -\frac{19}{17}P$$

$$\sum F_V = 0 \rightarrow R_A = 0$$